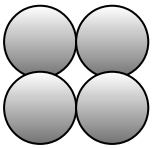
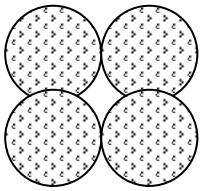
Year 9 Science

ELEMENTS, COMPOUNDS AND THEIR SYMBOLS

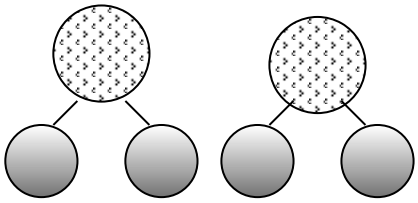
Elements are substances made up of all the same kinds of atoms. When different kinds of atoms join together, they form a molecule. Compounds are substances made up of the same kinds of molecules.



Hydrogen atoms form the element, hydrogen.



Oxygen atoms form the element, oxygen.



Two atoms of hydrogen combine with one atom of oxygen to form a water molecule. These water molecules form the compound, water.

The periodic table shows all the elements known to man. Each element is represented by a symbol. This symbol comes from the first one or two letters of the element's name. For example, carbon is C and helium is He. Sometimes the symbol is taken from the Latin or Arabic name of the element. For example, copper is Cu from its Latin name of cuprum.

If an element has a single-letter symbol, it is always written as a capital. Where the element has a two-letter symbol, only the first letter is written as a capital and the second letter is written in lower case.

Questions

1. Write the symbols for the following elements:

Aluminium _____	Magnesium _____	Sulfur _____
Copper _____	Nitrogen _____	Chlorine _____
Lead _____	Sodium _____	Iron _____
Nickel _____	Carbon _____	Mercury _____
Silver _____	Hydrogen _____	Silicon _____
Calcium _____	Manganese _____	Zinc _____
Gold _____	Oxygen _____	Potassium _____

2. Complete the following table:

COMPOUND	NUMBER AND TYPE OF ATOMS IN COMPOUND	FORMULA

Carbon dioxide	1 carbon, 2 oxygen	CO_2
Nitric acid	1 hydrogen, 1 nitrogen, 3 oxygen	
Copper sulfate		$CuSO_4$
Potassium permanganate		$KMnO_4$
Potassium dichromate		$K_2Cr_2O_7$
Sulfuric acid	2 hydrogen, 1 sulfur, 4 oxygen	
Sodium hydrogen carbonate		$NaHCO_3$
Copper carbonate	1 copper, 1 carbon, 3 oxygen	
Acetic acid		CH_3COOH
Sodium hydroxide	1 sodium, 1 oxygen, 1 hydrogen	

3. What elements do the following symbols represent?

Si _____
Hg _____
Au _____
B _____

Cl _____
F _____
Li _____
H _____

